

PROJECT STATEMENT

Student Attendance Management System

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1. Problem Statement

Managing student attendance manually can take time and may lead to calculation mistakes. Teachers or users need to keep track of the students who are present on each day and then calculate the total number of present days and attendance percentage for each student. The Student Attendance Management System provides a simple Python-based solution for recording attendance and automatically calculating attendance results.

2. Scope of the Project

The project focuses on basic student attendance management. The system allows the user to enter student names, specify the number of attendance days, record the students who are present on each day, and generate the final attendance result. The system calculates present days and attendance percentage automatically.

The current project is designed as a simple beginner-friendly system. Advanced features such as graphical user interfaces, database connectivity, online access, and automated notifications are outside the current scope but can be considered as future enhancements.

3. Target Users

- Teachers who need a simple way to record student attendance.
- Students who want to understand or demonstrate a basic attendance management system.
- Educational project evaluators who need to review the working of the system.

4. High-Level Features

- **Student Management:** Enter and store student names.
- **Attendance Management:** Record which students are present on each day.
- **Attendance Calculation:** Count the number of days each student is present.
- **Percentage Calculation:** Calculate attendance using present days and total days.
- **Attendance Report:** Display each student's name, present days, total days, and attendance percentage.
- **Basic Validation:** Identify names that are not present in the student list and avoid duplicate attendance for the same day.

5. Expected Outcome

The expected outcome is a simple and understandable attendance management program that reduces manual calculation work and produces an attendance result for every student. The project also demonstrates the use of basic Python programming concepts such as dictionaries, sets, loops, conditional statements, input/output, and arithmetic calculations.

6. Project Technology

The project is implemented using the Python programming language. The current implementation uses Python's built-in data structures and does not require an external database or framework.

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